

EMCH 260

SOLID MECHANICS

Ming Hu

Associate Professor
Department of Mechanical Engineering
University of South Carolina
E-Mail: hu@sc.edu

Stress

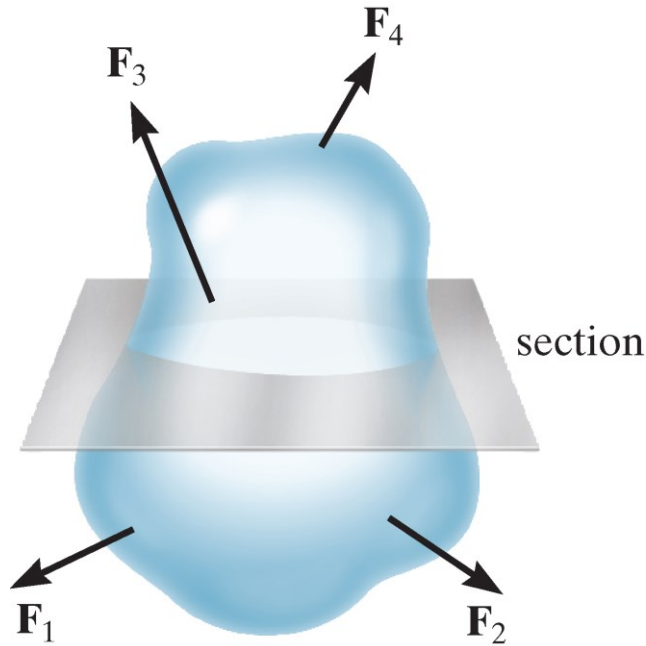
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Chapter Objectives

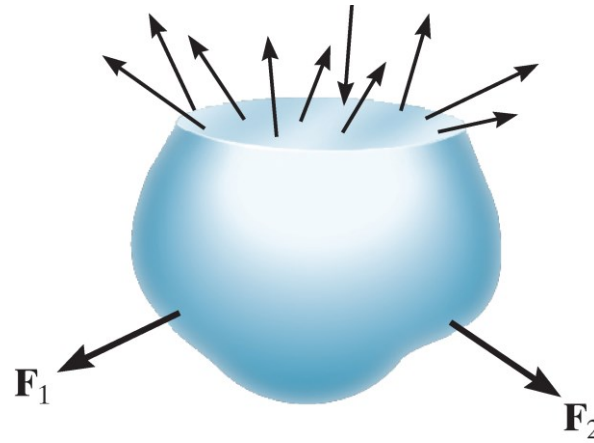
- ✓ Know how to write equations of equilibrium and solve the resultant internal loading
- ✓ Understand the concepts of normal and shear stress
- ✓ Analyze and design of members subjected on axial load or shear
- ✓ Know how to do allowable stress design

INTERNAL RESULTANT LOADINGS

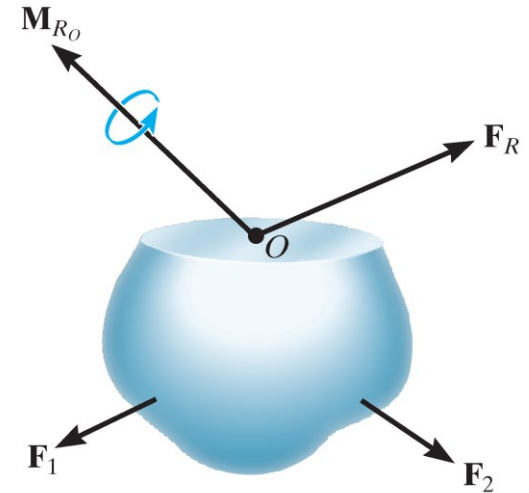
- Method of Sections



(a)



(b)



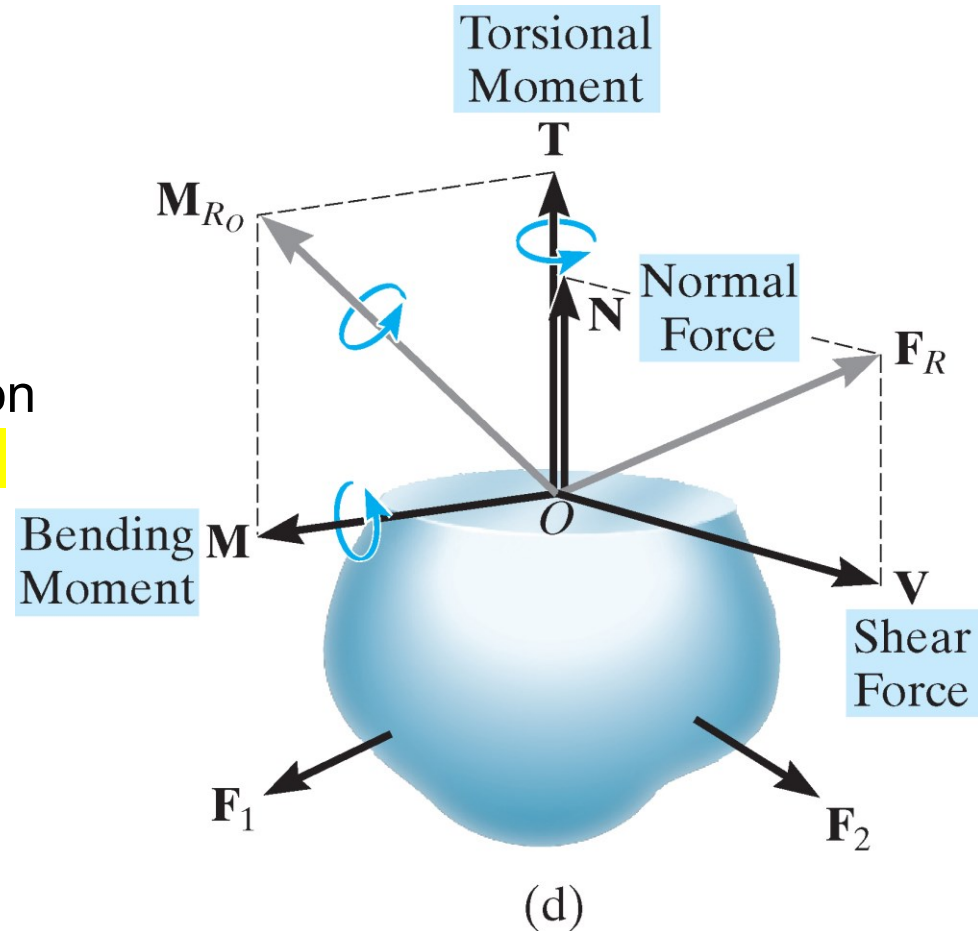
(c)

- Resultant loadings (**resultant effects of the distribution of loading**) [Fig. (c)], F_R and M_R , can be **known** by applying equations of equilibrium
- The imaginary section or “cut” through plane can be **arbitrary**, and the resultant loadings will also change accordingly

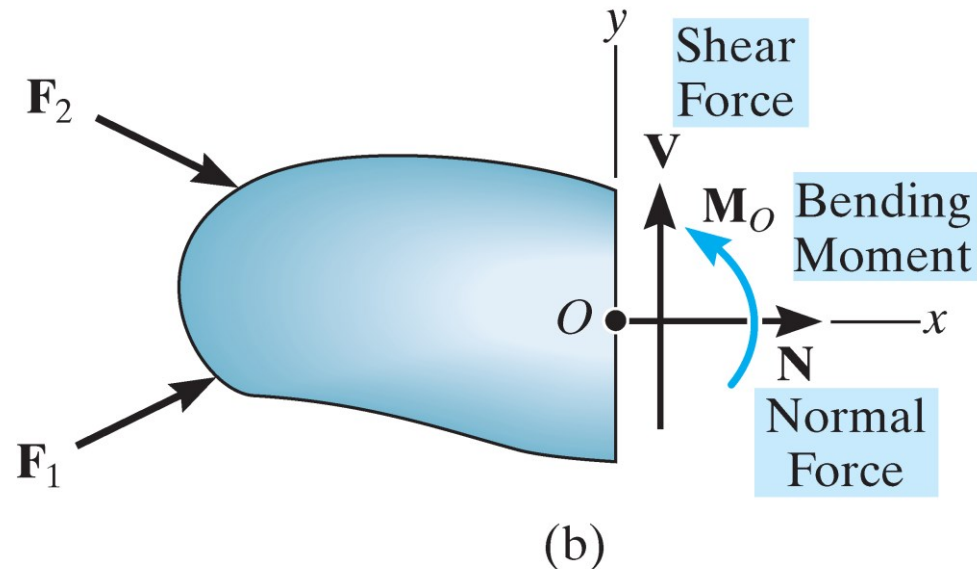
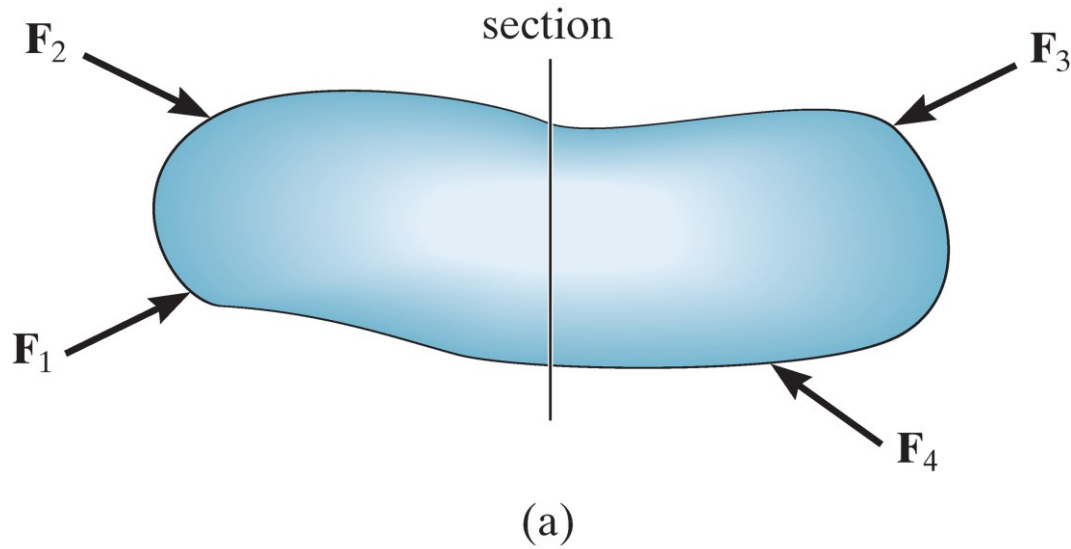
INTERNAL RESULTANT LOADINGS, CONT'D

- Types of resultant loadings (3D)

- **Normal force, N :** perpendicular to the section to prevent push or pull
- **Shear force, V :** in the section plane (relative movement of two segments)
- **Torsional moment or torque, T :** twist loads about an axis perpendicular to the section
- **Bending moment, M :** body bend about an axis lying within the plane of the section



COPLANNAR LOADINGS (xy plane)



- In case of coplanar system of forces, only **N**, **V** and **M** components exist (**WHY?**)

✓ Determine **N**:

$$\sum F_x = 0$$

✓ Determine **V**:

$$\sum F_y = 0$$

✓ Determine **M**:

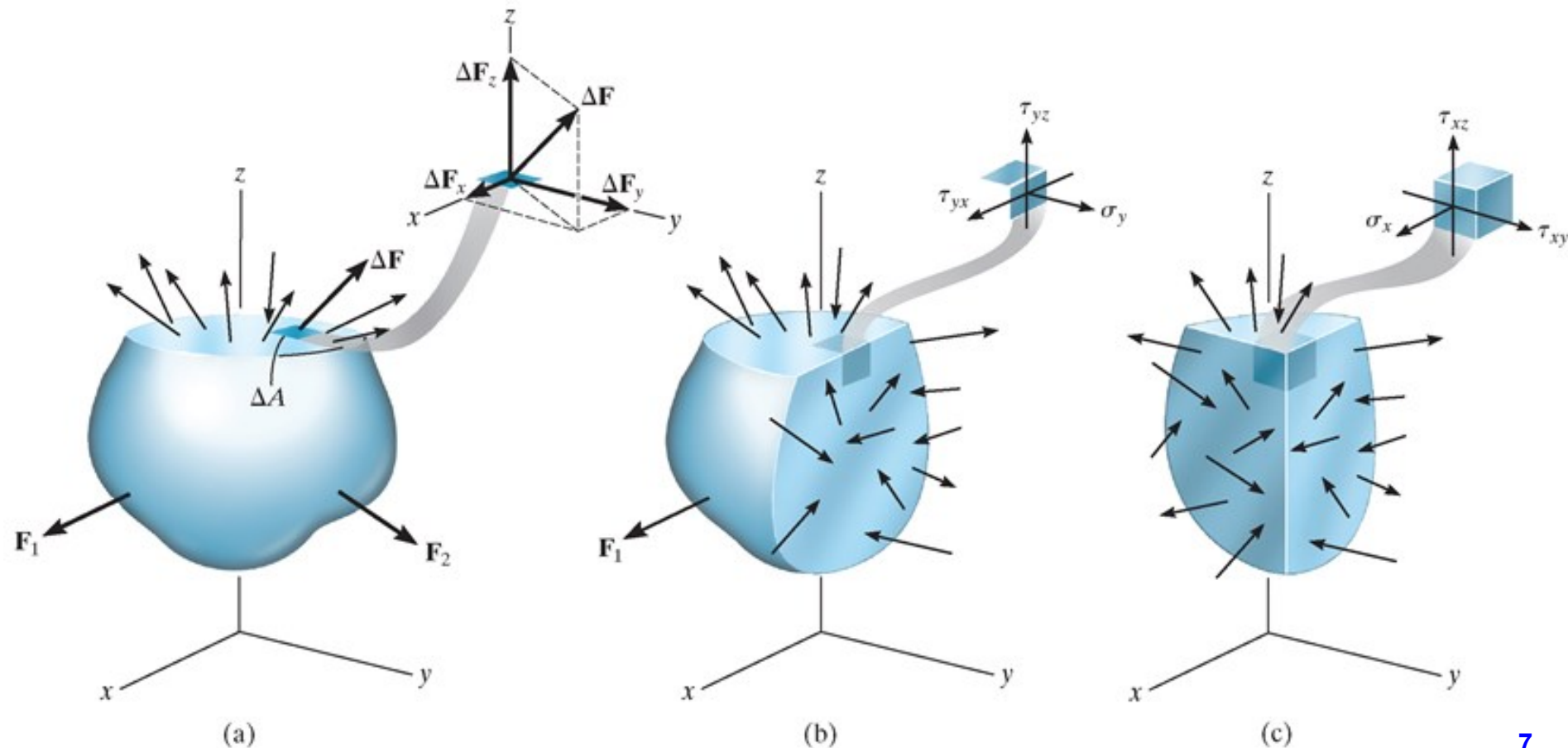
$$\sum M_z = 0$$

ANALYSIS PROCEDURE OF RESULTANT INTERNAL LOADINGS

- **Step 1: Support Reactions**
 - Determine the reactions acting on the sectioned segment if there are any or relevant (*sometimes the free-body diagram of the entire body should be considered*)
- **Step 2: Free-Body Diagram**
 - Draw free-body diagram for one of the sectioned segment and indicate the unknown resultants (**N**, **V**, **M**, and **T**)
 - For the case of coplanar system of forces, only **N**, **V**, and **M** are applied
- **Step 3: Equations of Equilibrium (EoEs)**
 - Determine unknown variables / quantities by applying EoEs.

STRESS – GENERAL DEFINITION

- For any macroscopically small area ΔA , there is a typical finite yet very small force ΔF . When using the continuum concept ($\Delta A \rightarrow 0$), $\Delta F \rightarrow 0$. But $\Delta F / \Delta A \rightarrow$ finite limit (*intensity of the internal force acting on a specific plane*).



NORMAL STRESS AND SHEAR STRESS

- Normal Stress (σ):** the force acting normal to ΔA

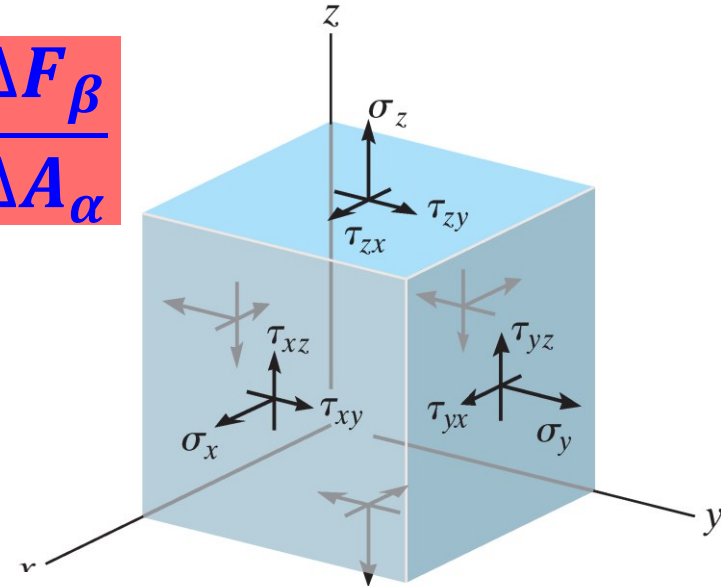
$$\sigma_x = \lim_{\Delta A \rightarrow 0} \frac{\Delta F_x}{\Delta A}$$

$$\sigma_y = \lim_{\Delta A \rightarrow 0} \frac{\Delta F_y}{\Delta A}$$

$$\sigma_z = \lim_{\Delta A \rightarrow 0} \frac{\Delta F_z}{\Delta A}$$

$$\sigma_{\alpha\beta} = \lim_{\Delta A_{\alpha} \rightarrow 0} \frac{\Delta F_{\beta}}{\Delta A_{\alpha}}$$

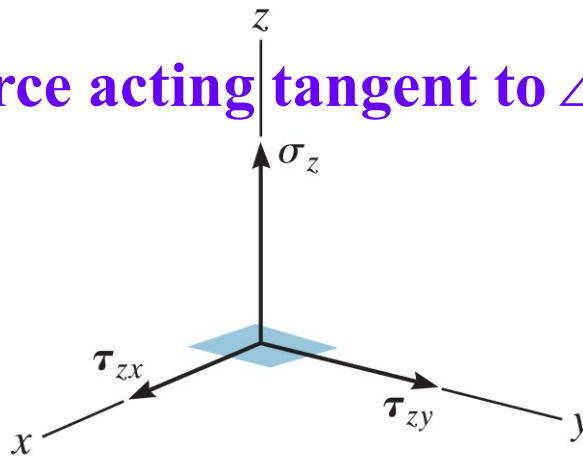
NOTE: ΔA is different for different normal stress



- Shear Stress (τ):** force acting tangent to ΔA

$$\tau_{zx} = \lim_{\Delta A_z \rightarrow 0} \frac{\Delta F_x}{\Delta A_z}$$

$$\tau_{zy} = \lim_{\Delta A_z \rightarrow 0} \frac{\Delta F_y}{\Delta A_z}$$



Units: N/m², Pa, pounds per square inch (psi)
(1 psi = 6.8947572932 kPa)

Chapter Objectives

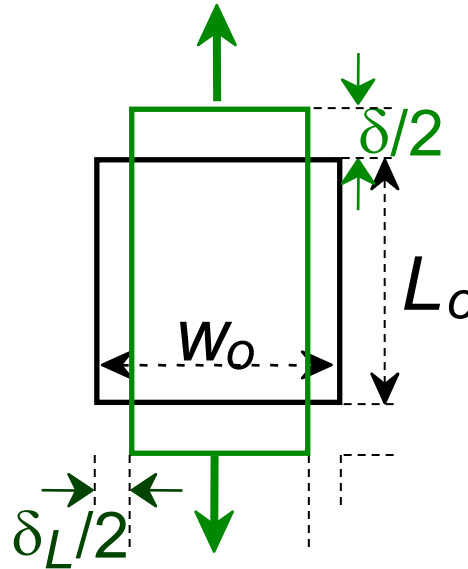
- ✓ Understand the concept of normal and shear strain
- ✓ Apply the concept to determine the strains for various types of problems

NORMAL STRAIN & SHEAR STRAIN

Normal strain (ϵ): measures the change in size (elongation/contraction)

- **Tensile strain:**

$$\epsilon = \frac{L - L_0}{L_0} = \frac{\delta}{L_0}$$



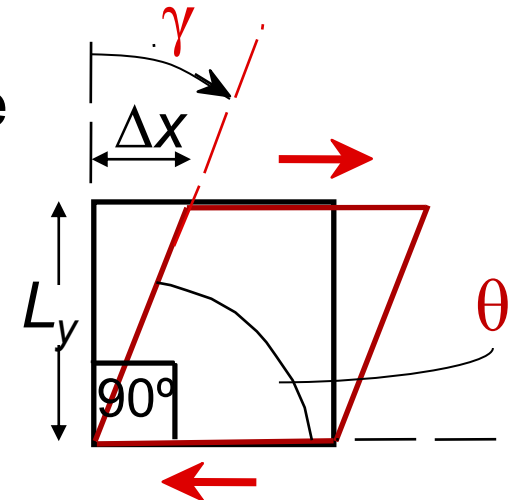
- **Lateral strain:**

$$\epsilon_L = \frac{w - w_0}{w_0} = \frac{\delta_L}{w_0}$$

Shearing strain (γ): measures the change in shape

- **Shear strain:**

$$\gamma = \frac{\pi}{2} - \theta$$



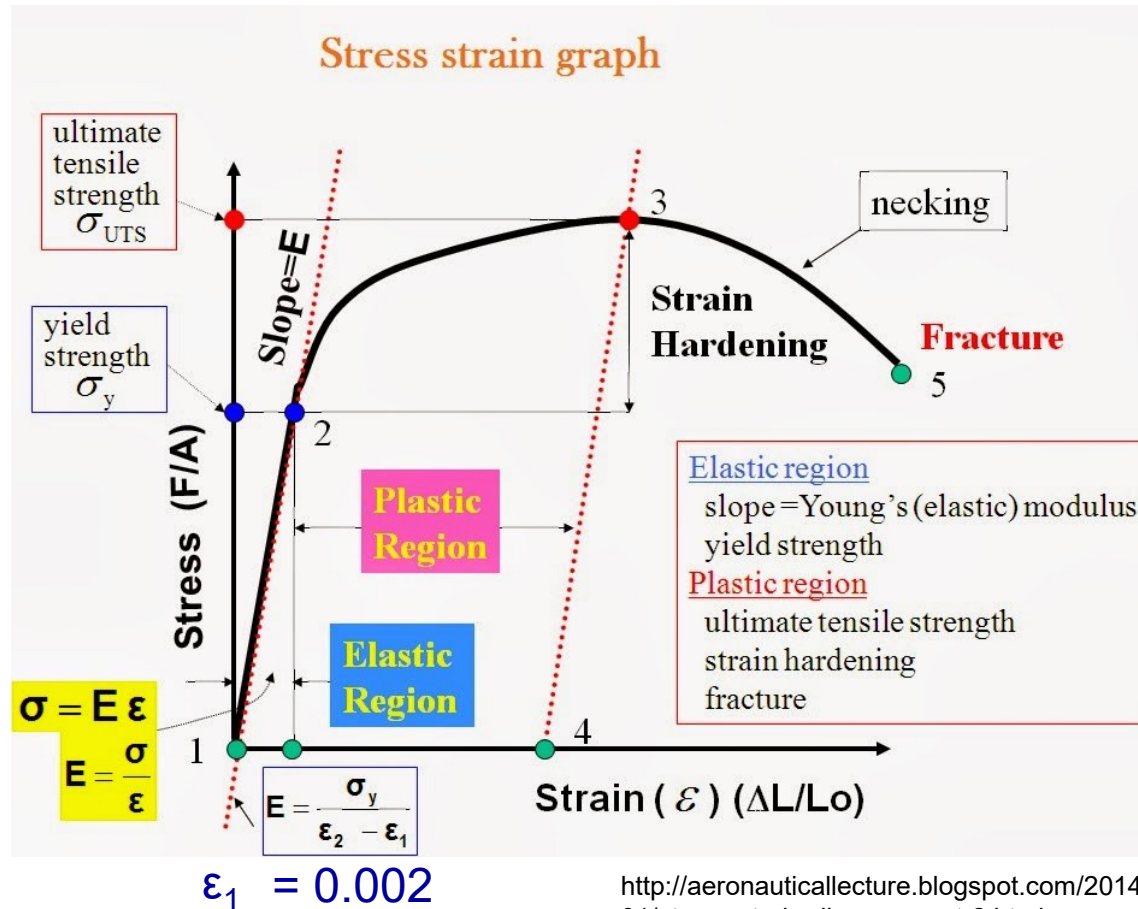
Mechanical Properties of Materials

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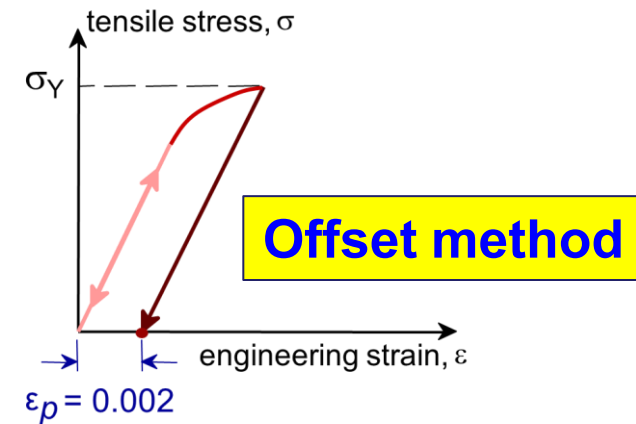
Chapter Objectives

- ✓ Understand stress-strain diagram
- ✓ Modulus of elasticity
- ✓ Yield strength
- ✓ Poisson's ratio
- ✓ Elastic and plastic deformation
- ✓ Idealized elastoplastic materials

STRESS-STRAIN DIAGRAM & HOOKE'S LAW



- Below the yield stress (σ_Y)
 $\sigma = E\varepsilon$
 E = Youngs Modulus or
 Modulus of Elasticity



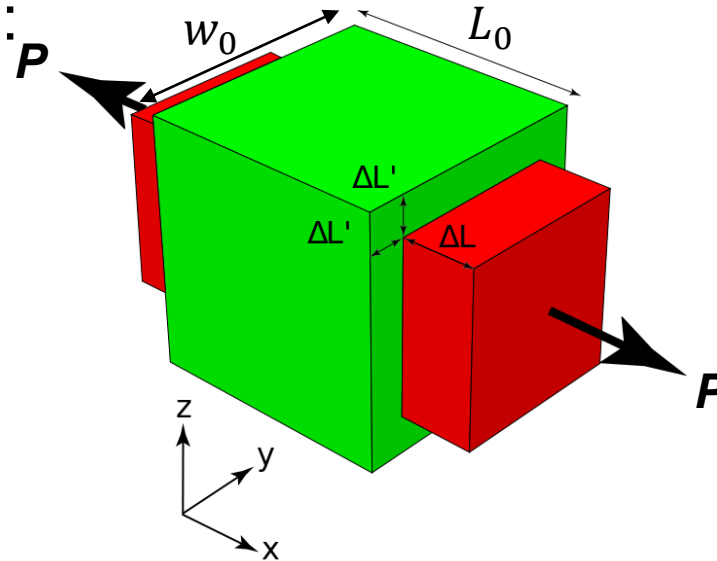
POISSON'S RATIO

Longitudinal strain:

$$\epsilon_{\text{long}} = \frac{\Delta L}{L_0}$$

Lateral strain:

$$\epsilon_{\text{lat}} = \frac{\Delta L'}{w_0}$$

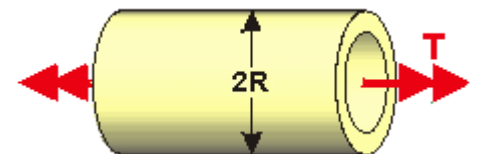
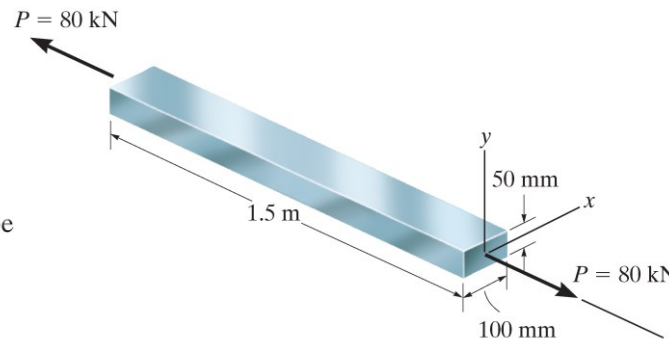
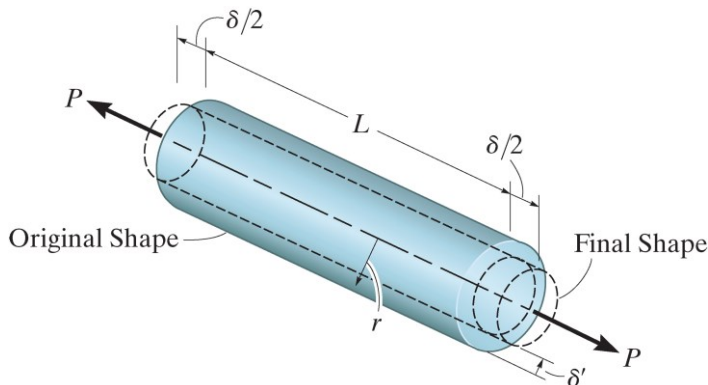


Green: before deformation
Red: after deformation

• **Poisson's Ratio**

$$\nu = - \frac{\epsilon_{\text{lat}}}{\epsilon_{\text{long}}}$$

- With the Poisson's ratio, one can know how the material will deform in the other (perpendicular) direction if the deformation in one direction is known.



IDEALIZED ELASTOPLASTIC MATERIAL

- The stress-strain diagram consists of the two straight-line segments

- When σ reaches the value σ_Y , the material starts yielding and keeps deforming plastically under a constant load.

- If the load is removed, unloading takes place along a straight-line segment CD parallel to the initial portion AY of the loading curve.

- The total strain is equal to the recovered part (ϵ_{el}) plus the permanent set part (ϵ_{pl})

